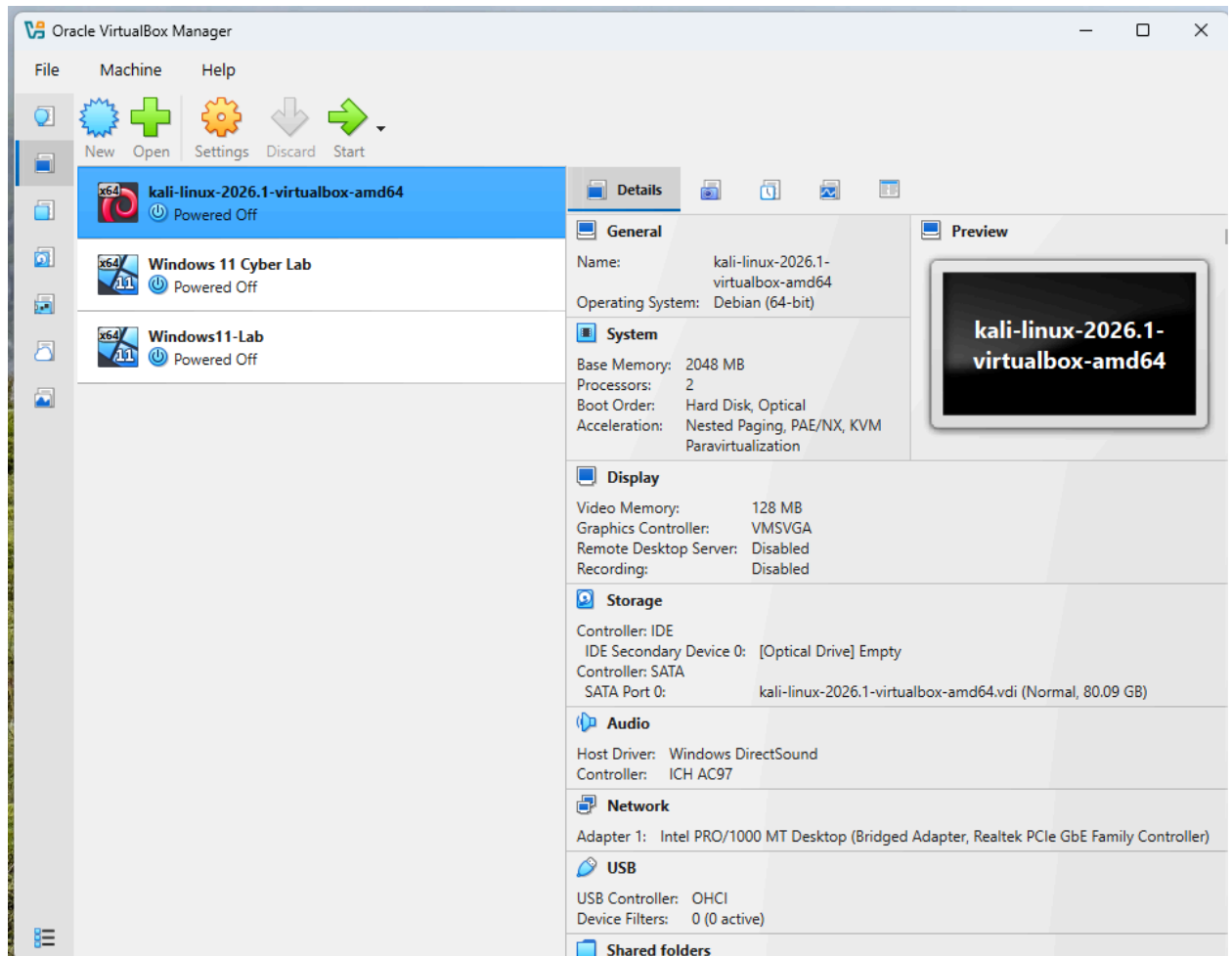
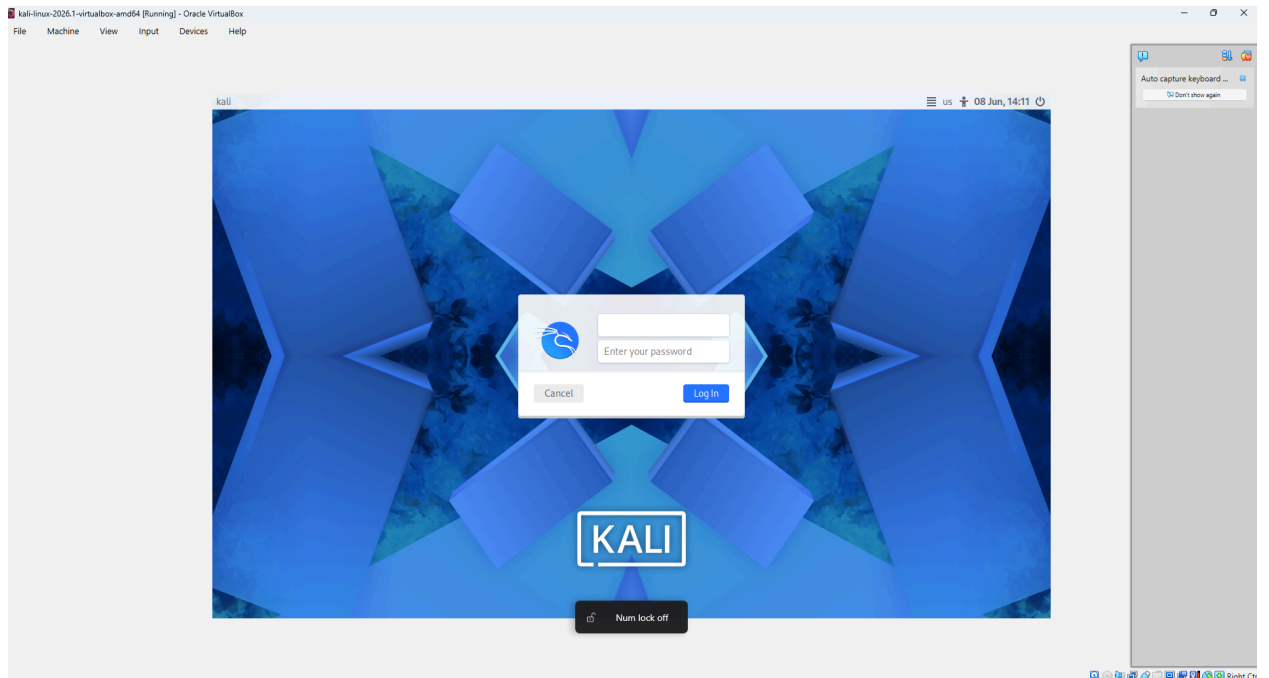
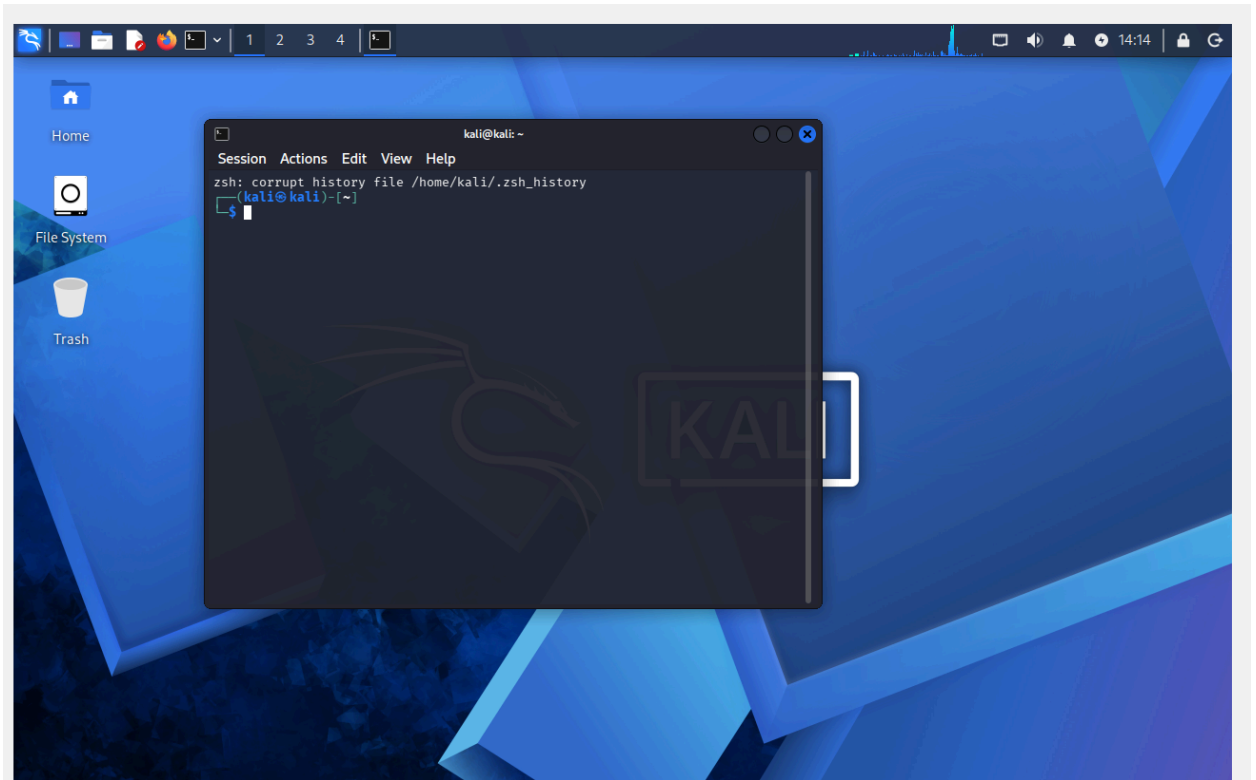


Built a home cybersecurity lab using Kali Linux and Windows. Performed host discovery with Nmap, packet analysis with Wireshark, Windows Event Log investigation, PowerShell-based threat hunting, service enumeration, patch auditing, and network connection analysis.







Get-Process | Sort-Object CPU -Descending | Select-Object -First 15

What you're looking for

Process Name
CPU Usage
Memory Usage

Question

Which process is consuming the most resources?

Resume Skill

Investigated running processes using PowerShell.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\System32> Get-process | Sort-Object CPU -Descending | Select-Object -First 15

Handles NPM(K) PM(K) WS(K) CPU(s) Id SI ProcessName
-----
1292 48 142900 164448 61.56 308 5 XboxPcApp
397 37 340416 417164 46.11 30900 5 chrome
1244 142 190928 81864 38.33 12764 5 EpicGamesLauncher
2254 80 115532 274556 34.97 24224 5 chrome
3244 101 133760 209276 27.91 10576 5 explorer
1918 53 190844 188776 24.86 15268 5 chrome
1891 62 68484 104496 21.00 24852 5 wavebrowser
973 86 81028 21424 17.92 19340 5 Adobe Desktop Service
341 45 215376 11796 15.64 15964 5 msedgewebview2
1233 43 39736 84112 10.33 31596 5 msedgewebview2
1096 50 55628 103624 9.14 13336 5 SearchHost
391 32 90752 136952 8.08 17424 5 wavebrowser
575 31 14188 27032 7.89 34944 5 uihost
1491 76 162328 81400 7.81 12544 5 RadeonSoftware
1195 55 74516 100816 7.66 20408 5 StartMenuExperienceHost

PS C:\WINDOWS\System32>
```

Get-NetTCPConnection | Where-Object {\$_.State -eq "Established"}

What you're looking for

Remote IP
Remote Port
Local Port
State

Question

What external IPs is this computer talking to?

```
PS C:\WINDOWS\System32> Get-NetTCPConnection | Where-Object { $_.State -eq "Established" }

LocalAddress LocalPort RemoteAddress RemotePort State AppliedSetting OwningProcess
-----
2603:900b:300:61bc:f953:a583:1cc... 63512 2600:1403:9c00:d94::57 443 Established Internet 4064
2603:900b:300:61bc:f953:a583:1cc... 59840 2607:f8b0:400c:c01::bc 5228 Established Internet 12804
2603:900b:300:61bc:f953:a583:1cc... 59431 2a06:98c1:3108::ac40:94eb 443 Established Internet 11472
2603:900b:300:61bc:f953:a583:1cc... 58050 2607:f8b0:400c:c00::bc 5228 Established Internet 11472
2603:900b:300:61bc:f953:a583:1cc... 52326 2001:4998:3c:807::2 443 Established Internet 12804
2603:900b:300:61bc:f953:a583:1cc... 52325 2607:f8b0:4008:811::2005 443 Established Internet 12804
2603:900b:300:61bc:f953:a583:1cc... 52324 2607:f8b0:4008:811::2005 443 Established Internet 12804
2603:900b:300:61bc:f953:a583:1cc... 52323 2606:4700:4406::6812:2bcc 443 Established Internet 11472
2603:900b:300:61bc:f953:a583:1cc... 51116 2a06:98c1:310b::ac40:9bd1 443 Established Internet 11472
192.168.1.161 65376 192.168.1.120 8009 Established Internet 11472
192.168.1.161 61006 172.183.7.194 443 Established Internet 5376
192.168.1.161 59831 192.168.1.120 8009 Established Internet 12804
192.168.1.161 57179 3.210.231.143 443 Established Internet 3432
127.0.0.1 52365 127.0.0.1 52363 Established Internet 5284
127.0.0.1 52363 127.0.0.1 52365 Established Internet 25692

PS C:\WINDOWS\System32>
```

Lab 3: Find Local Administrators

Goal

Determine who has admin rights.

Command

Get-LocalGroupMember Administrators

What you're looking for

Unexpected users

Unknown accounts

Why hackers do this

Attackers often check:

Who has admin privileges?

after gaining access.

```
PS C:\WINDOWS\System32> Get-LocalGroupMember Administrators

ObjectClass Name                PrincipalSource
-----
User         Homeoffice\Administrator Local
User         HOMEOFFICE\ALYAS         MicrosoftAccount
```

Lab 5: Find Installed Programs

Goal

Enumerate installed software.

Command

Get-Package

What you're looking for

Unknown software

Old software

Potentially vulnerable software

Why hackers do this

Attackers look for:

Outdated software

Known vulnerabilities

```
PS C:\WINDOWS\System32> Get-Package
```

Name	Version	Source	ProviderName
thinkorswim	desktop		Programs
HP Documentation	1.0.0.1		Programs
Office 16 Click-to-Run Ext...	16.0.20026.20076		msi
Microsoft 365 - en-us	16.0.20026.20140		Programs
Microsoft OneNote - en-us	16.0.20026.20140		Programs
Office 16 Click-to-Run Loca...	16.0.14026.20302		msi
Microsoft GameInput	3.3.221.0		msi
Microsoft Visual C++ 2022 X...	14.44.35211		msi
Microsoft Visual C++ 2013 x...	12.0.40664		msi
Microsoft Visual C++ 2022 X...	14.44.35211		msi
Oracle VirtualBox 7.2.8	7.2.8		msi
TikTok LIVE Studio	1.25.2		Programs
Microsoft Visual C++ 2013 x...	12.0.40664		msi
Epic Games Launcher Prerequ...	1.0.0.0		msi
Microsoft Visual C++ 2022 X...	14.44.35211		msi
ExpressVPN	10.0.9.2		msi
ExpressVPN	10.0.9.2		Programs
Microsoft Visual C++ 2012 x...	11.0.61030		msi
Microsoft Visual C++ 2022 X...	14.44.35211		msi
Kraken Desktop	1.17.2		msi
Microsoft Visual C++ 2013 x...	12.0.40664		msi
Epic Online Services	2.0.44.0		msi
Microsoft Visual C++ 2012 x...	11.0.61030		msi
Microsoft Visual C++ 2012 x...	11.0.61030		msi
HP One Agent	1.3.214.7339	C:\Program Files\HP\HP One Ag...	msi
HP One Agent	1.3.214.7339		Programs
HP PC Hardware Diagnostics ...	9.1.0.0	C:\Windows\HP\	msi
Trader Workstation	latest (10.47...		Programs
Microsoft Visual C++ 2013 x...	12.0.40664		msi
Power E*TRADE Desktop 148.0.0	148.0.0		Programs
HP Audio Switch	1.0.211.0		msi
DCS FB Marketplace Posting ...	1.0.0.108		Programs
Microsoft Visual C++ 2012 x...	11.0.61030		msi
Microsoft OneDrive	26.088.0510.0004		Programs
Microsoft Update Health Tools	5.72.0.0		msi
Roblox Player for ALYAS			Programs
Roblox Studio for ALYAS			Programs
TC2000 v20	1.0.0		Programs
TC2000 v20	1.0.0	C:\Users\ALYAS\AppData\Roamin...	msi
Epic Games Launcher	1.3.93.0	C:\Program Files (x86)\Epic G...	msi
Wave Browser	1.5.24.14		Programs
Adobe Creative Cloud	6.9.1.1		Programs
Browser Extension	1.20.105.0	C:\Users\ALYAS\AppData\Roamin...	msi
Adobe Genuine Service	9.1.0.52		Programs
Microsoft Teams Meeting Add...	1.26.11802		msi
Google Chrome	149.0.7827.53		Programs
BBWC	1.21.1026.21622	C:\Users\ALYAS\AppData\Roamin...	msi
Copilot	148.0.3967.96		Programs
Microsoft Edge	149.0.4022.52		Programs
McAfee LiveSafe	16.0 R43		Programs
Adobe Photoshop 2023	24.3.0.376		Programs
UXP WebView Support	1.1.0		Programs
WildTangent Games	1.1.1.71		Programs

Run:

Get-ComputerInfo

Why attackers do this

One of the first things attackers do after gaining access is:

Discover:

- OS version
- Computer name
- Installed patches
- Domain membership

This is called **host enumeration**.

```
PS C:\WINDOWS\System32> Get-ComputerInfo

WindowsBuildLabEx           : 26100.1.amd64fre.ge_release.240331-1435
WindowsCurrentVersion       : 6.3
WindowsEditionId            : Core
WindowsInstallationType     : Client
WindowsInstallDateFromRegistry : 2/22/2026 11:05:54 PM
WindowsProductId            : 00342-20760-08030-AAOEM
WindowsProductName          : Windows 10 Home
WindowsRegisteredOrganization : HP
WindowsRegisteredOwner      : ALYASABADY@outlook.com
WindowsSystemRoot            : C:\WINDOWS
WindowsVersion               : 20H2
OSDisplayVersion             : 25H2
BiosCharacteristics          : {7, 9, 11, 12...}
BiosBIOSVersion              : {HPQOEM - 1072009, F.21, American Megatrends - 50011}
BiosBuildNumber              :
BiosCaption                  : F.21
BiosCodeSet                  :
BiosCurrentLanguage          : en|US|iso8859-1
BiosDescription              : F.21
BiosEmbeddedControllerMajorVersion : 103
BiosEmbeddedControllerMinorVersion : 17
BiosFirmwareType             : Uefi
BiosIdentificationCode       :
BiosInstallableLanguages     : 13
BiosInstallDate              :
BiosLanguageEdition          :
BiosListOfLanguages          : {en|US|iso8859-1, fr|FR|iso8859-1, es|ES|iso8859-1, de|DE|iso8859-1...}
BiosManufacturer             : AMI
BiosName                     : F.21
BiosOtherTargetOS            :
BiosPrimaryBIOS              : True
BiosReleaseDate              : 9/16/2021 8:00:00 PM
BiosSerialNumber              : 2M014832PM
BiosSMBIOSBIOSVersion        : F.21
BiosSMBIOSMajorVersion       : 3
BiosSMBIOSMinorVersion       : 3
BiosSMBIOSPresent            : True
BiosSoftwareElementState     : Running
BiosStatus                   : OK
BiosSystemBiosMajorVersion    : 15
BiosSystemBiosMinorVersion    : 33
BiosTargetOperatingSystem     : 0
BiosVersion                   : HPQOEM - 1072009
CsAdminPasswordStatus         : Unknown
CsAutomaticManagedPagefile   : True
CsAutomaticResetBootOption   : True
CsAutomaticResetCapability    : True
CsBootOptionOnLimit           :
CsBootOptionOnWatchDog        :
CsBootROMSupported            : True
CsBootStatus                  : {0, 0, 0, 0...}
```

Challenge

Run these three commands:

Get-Process | Sort-Object CPU -Descending | Select-Object -First 10

Get-NetTCPConnection | Where-Object {\$_.State -eq "Established"}

Get-LocalGroupMember Administrators

1. Process Analysis

Top CPU consumers:

chrome
chrome
chrome
XboxPcApp
explorer
EpicGamesLauncher
WaveBrowser
SnippingTool
Adobe Desktop Service

What I'd investigate

✅ Chrome, Explorer, SnippingTool, Adobe are normal.

⚠️ **WaveBrowser** is worth investigating.

Many SOC analysts flag Wave Browser because it has historically been associated with unwanted software and adware behavior.

```
PS C:\WINDOWS\System32> Get-Process | Sort-Object CPU -Descending | Select-Object -First 10
```

Handles	NPM(K)	PM(K)	WS(K)	CPU(s)	Id	SI	ProcessName
401	36	374556	394888	157.92	30900	5	chrome
1282	47	151716	96716	93.22	308	5	XboxPcApp
2321	83	129340	257324	81.39	24224	5	chrome
1938	56	345340	212728	63.66	15268	5	chrome
3352	102	134884	200260	49.83	10576	5	explorer
1244	142	190936	27556	43.80	12764	5	EpicGamesLauncher
580	71	301816	406684	39.69	34784	5	chrome
1846	61	68368	95432	28.78	24852	5	wavebrowser
1355	63	340420	394064	20.56	22660	5	SnippingTool
979	86	80808	17824	19.34	19340	5	Adobe Desktop Service

```
PS C:\WINDOWS\System32>
PS C:\WINDOWS\System32> Get-NetTCPConnection | Where-Object {$_.State -eq "Established"}
```

LocalAddress	LocalPort	RemoteAddress	RemotePort	State	AppliedSetting	OwningProcess
2603:900b:300:61bc:f953:a583:1cc...	59840	2607:f8b0:400c:c01::bc	5228	Established	Internet	12804
2603:900b:300:61bc:f953:a583:1cc...	59431	2a06:98c1:3108::ac40:94eb	443	Established	Internet	11472
2603:900b:300:61bc:f953:a583:1cc...	58050	2607:f8b0:400c:c00::bc	5228	Established	Internet	11472
2603:900b:300:61bc:f953:a583:1cc...	56549	2406:da00:ff00::3d0:ac5d	443	Established	Internet	3432
2603:900b:300:61bc:f953:a583:1cc...	56548	2001:4098:3c:807::1	443	Established	Internet	12804
2603:900b:300:61bc:f953:a583:1cc...	56547	2607:f8b0:4008:811::2005	443	Established	Internet	12804
2603:900b:300:61bc:f953:a583:1cc...	51116	2a06:98c1:310b::ac40:9bd1	443	Established	Internet	11472
192.168.1.161	65376	192.168.1.120	8009	Established	Internet	11472
192.168.1.161	61006	172.183.7.194	443	Established	Internet	5376
192.168.1.161	59831	192.168.1.120	8009	Established	Internet	12804
127.0.0.1	52365	127.0.0.1	52363	Established	Internet	5284
127.0.0.1	52363	127.0.0.1	52365	Established	Internet	25692

```
PS C:\WINDOWS\System32>
PS C:\WINDOWS\System32> Get-LocalGroupMember Administrators
```

ObjectClass	Name	PrincipalSource
User	Homeoffice\Administrator	Local
User	HOMEOFFICE\ALYAS	MicrosoftAccount

Another Great Lab

Find Startup Programs

Run:

Get-CimInstance Win32_StartupCommand

Look for:

Adobe
 Chrome
 OneDrive
 Epic Games
 Unknown software

This is one of the first things malware analysts check.

```
PS C:\WINDOWS\System32> Get-Item "C:\Program Files\WaveBrowser\*" -ErrorAction SilentlyContinue
PS C:\WINDOWS\System32> Get-CimInstance Win32_StartupCommand

Command
-----
C:\Program Files\HP\HP System Event Utility\Host Launcher\HpseuHostLauncher.exe
C:\Users\ALYAS\AppData\Local\Microsoft\OneDrive\OneDrive.exe /background
289
C:\Program Files (x86)\Epic Games\Launcher\Portal\Binaries\Win64\EpicGamesLauncher.exe -silent -launchcontext-boot
C:\Users\ALYAS\AppData\Local\Microsoft\WindowsApps\MicrosoftTeams.exe --no-startup-window --win-session-start
C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe --no-startup-window --win-session-start
windir\system32\SecurityHealthSystray.exe
C:\Program Files (x86)\Common Files\Adobe\AdobeGCClient\AGCInvokerUtility.exe"

User      Caption
-----
HOMEOFFICE\ALYAS HPSEU_Host_Launcher
HOMEOFFICE\ALYAS OneDrive
HOMEOFFICE\ALYAS
HOMEOFFICE\ALYAS EpicGamesLauncher
HOMEOFFICE\ALYAS Teams
HOMEOFFICE\ALYAS MicrosoftEdgeAutoLaunch_CED9008A756206816A92862C7F8D638C
Public SecurityHealth
Public AdobeGCInvoker-1.0

PS C:\WINDOWS\System32>
```

Challenge

Run this command:

Get-NetTCPConnection | Sort-Object RemoteAddress | Select-Object -First 20

Pick one IP you don't recognize and we'll investigate it like a Tier 1 SOC analyst:

- Who owns it?
- What service is it?
- Is it suspicious?
- Is it normal traffic?

```
PS C:\WINDOWS\System32> Get-NetTCPConnection | Sort-Object RemoteAddress | Select-Object -First 20
```

LocalAddress	LocalPort	RemoteAddress	RemotePort	State	AppliedSetting	OwningProcess
::	49673	::	0	Listen		1164
::	49668	::	0	Listen		4708
::	49667	::	0	Listen		2584
::	49463	::	0	Bound		5456
::	52358	::	0	Bound		5456
::	50034	::	0	Bound		4064
::	49782	::	0	Bound		4064
::	7680	::	0	Listen		6936
::	445	::	0	Listen		4
::	135	::	0	Listen		1508
::1	35783	::	0	Listen		3028
::	49666	::	0	Listen		836
::	49665	::	0	Listen		1036
::	49664	::	0	Listen		1196
::	59431	::	0	Bound		11472
::	59043	::	0	Bound		308
::	61066	::	0	Bound		35828
::	59500	::	0	Bound		4064
::	58546	::	0	Bound		2392
::	58356	::	0	Bound		4064

Which process owns these ports?

Cybersecurity Lab #1 — Find Listening Ports

Run:

Get-NetTCPConnection -State Listen

```
PS C:\WINDOWS\System32> Get-Process -Id 4,1508,6936
```

Handles	NPM(K)	PM(K)	WS(K)	CPU(s)	Id	SI	ProcessName
1812	19	17624	19268		1508	0	svchost
522	20	11336	14944		6936	0	svchost
6394	0	52	1100		4	0	System


```
PS C:\WINDOWS\System32> Get-NetTCPConnection -State Listen
```

LocalAddress	LocalPort	RemoteAddress	RemotePort	State	AppliedSetting	OwningProcess
::	49673	::	0	Listen		1164
::	49668	::	0	Listen		4708
::	49667	::	0	Listen		2584
::	49666	::	0	Listen		836
::	49665	::	0	Listen		1036
::	49664	::	0	Listen		1196
:::1	35783	::	0	Listen		3028
::	7680	::	0	Listen		6936
::	445	::	0	Listen		4
::	135	::	0	Listen		1508
127.0.0.1	52363	0.0.0.0	0	Listen		25692
0.0.0.0	49673	0.0.0.0	0	Listen		1164
0.0.0.0	49668	0.0.0.0	0	Listen		4708
0.0.0.0	49667	0.0.0.0	0	Listen		2584
0.0.0.0	49666	0.0.0.0	0	Listen		836
0.0.0.0	49665	0.0.0.0	0	Listen		1036
0.0.0.0	49664	0.0.0.0	0	Listen		1196
127.0.0.1	15292	0.0.0.0	0	Listen		19340
0.0.0.0	6646	0.0.0.0	0	Listen		7148
0.0.0.0	5040	0.0.0.0	0	Listen		11120
127.0.0.1	2015	0.0.0.0	0	Listen		5284
192.168.56.1	139	0.0.0.0	0	Listen		4
192.168.1.161	139	0.0.0.0	0	Listen		4
0.0.0.0	135	0.0.0.0	0	Listen		1508

Then:

Get-NetTCPConnection -State Listen | Sort-Object LocalPort

Look for unusual ports such as:

- 3389 (Remote Desktop)
- 445 (SMB File Sharing)
- 135 (RPC)
- 5985 (PowerShell Remoting)

These are common ports attackers and defenders care about.

```
PS C:\WINDOWS\System32> Get-NetTCPConnection -State Listen | Sort-Object LocalPort
```

LocalAddress	LocalPort	RemoteAddress	RemotePort	State	AppliedSetting	OwningProcess
::	135	::	0	Listen		1508
0.0.0.0	135	0.0.0.0	0	Listen		1508
192.168.1.161	139	0.0.0.0	0	Listen		4
192.168.56.1	139	0.0.0.0	0	Listen		4
::	445	::	0	Listen		4
127.0.0.1	2015	0.0.0.0	0	Listen		5284
0.0.0.0	5040	0.0.0.0	0	Listen		11120
0.0.0.0	6646	0.0.0.0	0	Listen		7148
::	7680	::	0	Listen		6936
127.0.0.1	15292	0.0.0.0	0	Listen		19340
::1	35783	::	0	Listen		3028
::	49664	::	0	Listen		1196
0.0.0.0	49664	0.0.0.0	0	Listen		1196
0.0.0.0	49665	0.0.0.0	0	Listen		1036
::	49665	::	0	Listen		1036
::	49666	::	0	Listen		836
0.0.0.0	49666	0.0.0.0	0	Listen		836
0.0.0.0	49667	0.0.0.0	0	Listen		2584
::	49667	::	0	Listen		2584
::	49668	::	0	Listen		4708
0.0.0.0	49668	0.0.0.0	0	Listen		4708
::	49673	::	0	Listen		1164
0.0.0.0	49673	0.0.0.0	0	Listen		1164
127.0.0.1	52363	0.0.0.0	0	Listen		25692

Cybersecurity Lab #2 — Find Startup Programs

Run:

```
Get-CimInstance Win32_StartupCommand | Select-Object Name, Command
```

This is one of the first places malware analysts check.

You're looking for:

Unknown executables

Random names

Programs launching from Temp folders

```
PS C:\WINDOWS\System32> Get-Service | Where-Object {$_.Status -eq "Running"}
```

Status	Name	DisplayName
-----	----	-----
Running	AdobeUpdateService	AdobeUpdateService
Running	AGMSvc	Adobe Genuine Software Monitor Service
Running	AMD Crash Defen...	AMD Crash Defender Service
Running	AMD External Ev...	AMD External Events Utility
Running	Appinfo	Application Information
Running	AppXSvc	AppX Deployment Service (AppXSVC)
Running	AudioEndpointBu...	Windows Audio Endpoint Builder
Running	Audiosrv	Windows Audio
Running	BFE	Base Filtering Engine
Running	BITS	Background Intelligent Transfer Ser...
Running	BluetoothUserSe...	Bluetooth User Support Service_ab374af
Running	BrokerInfrastru...	Background Tasks Infrastructure Ser...
Running	BTAGService	Bluetooth Audio Gateway Service
Running	BthAvctpSvc	AVCTP service
Running	bthserv	Bluetooth Support Service
Running	camsvc	Capability Access Manager Service
Running	cbdhsvc_ab374af	Clipboard User Service_ab374af
Running	CDPSvc	Connected Devices Platform Service
Running	CDPUserSvc_ab374af	Connected Devices Platform User Ser...
Running	ClickToRunSvc	Microsoft Office Click-to-Run Service
Running	CoreMessagingRe...	CoreMessaging
Running	CryptSvc	Cryptographic Services
Running	DcomLaunch	DCOM Server Process Launcher
Running	DeviceAssociati...	Device Association Service
Running	DeviceInstall	Device Install Service
Running	DevicesFlowUser...	DevicesFlow_ab374af
Running	DevQueryBroker	DevQuery Background Discovery Broker
Running	Dhcp	DHCP Client
Running	DiagTrack	Connected User Experiences and Tele...
Running	DispBrokerDeskt...	Display Policy Service
Running	DisplayEnhancem...	Display Enhancement Service
Running	Dnscache	DNS Client
Running	DoSvc	Delivery Optimization
Running	DPS	Diagnostic Policy Service
Running	DsSvc	Data Sharing Service
Running	DusmSvc	Data Usage
Running	EventLog	Windows Event Log
Running	EventSystem	COM+ Event System
Running	ExpressVPNService	ExpressVPN Service
Running	FontCache	Windows Font Cache Service
Running	GameInputRedist...	GameInput Redist Service
Running	GameInputSvc	GameInput Service
Running	GamingServices	Gaming Services
Running	GamingServicesNet	Gaming Services
Running	hidserv	Human Interface Device Service
Running	HP Comm Recover	HP Comm Recovery
Running	HPAppHelperCap	HP App Helper HSA Service
Running	HPDiagsCap	HP Diagnostics HSA Service
Running	HPNetworkCap	HP Network HSA Service
Running	hp-one-agent-se...	HP One Agent Service
Running	HPPrintScanDoct...	HP Print Scan Doctor Service
Running	HPSysInfoCap	HP System Info HSA Service
Running	HpTouchpointAna...	HP Insights Analytics
Running	ID19 HP Fan Con...	ID19 HP Fan Control Service

```

Running InstallService Microsoft Store Install Service
Running InventorySvc Inventory and Compatibility Apprais...
Running iphlpsvc IP Helper
Running KeyIso CNG Key Isolation
Running LanmanServer Server
Running LanmanWorkstation Workstation
Running lfsvc Geolocation Service
Running LicenseManager Windows License Manager Service
Running lmhosts TCP/IP NetBIOS Helper
Running LSM Local Session Manager
Running McAfee WebAdvisor McAfee WebAdvisor
Running McAPExe McAfee AP Service
Running mcccspvc McAfee CSP Service
Running MDCoreSvc Microsoft Defender Core Service
Running mfemms McAfee Service Controller
Running mfevtp McAfee Validation Trust Protection ...
Running ModuleCoreService McAfee Module Core Service
Running mpssvc Windows Defender Firewall
Running NcbService Network Connection Broker
Running Netman Network Connections
Running netprofm Network List Service
Running NgcCtnrSvc Microsoft Passport Container
Running NgcSvc Microsoft Passport
Running NPSMSvc_ab374af Now Playing Session Manager Service...
Running nsi Network Store Interface Service
Running PcaSvc Program Compatibility Assistant Ser...
Running PEFService McAfee PEF Service
Running PimIndexMainten... Contact Data_ab374af
Running PlugPlay Plug and Play
Running Power Power
Running ProfSvc User Profile Service
Running QWAVE Quality Windows Audio Video Experience
Running RasMan Remote Access Connection Manager
Running RmSvc Radio Management Service
Running RpcEptMapper RPC Endpoint Mapper
Running RpcSs Remote Procedure Call (RPC)
Running RtkAudioUnivers... Realtek Audio Universal Service
Running RtkBtManServ Realtek Bluetooth Device Manager Se...
Running SamSs Security Accounts Manager
Running Schedule Task Scheduler
Running seclogon Secondary Logon
Running SECOMNService Sound Research SECOMN Service
Running SecurityHealthS... Windows Security Service
Running SENS System Event Notification Service
Running ShellHWDetection Shell Hardware Detection
Running Spooler Print Spooler
Running SSDPSRV SSDP Discovery
Running SstpSvc Secure Socket Tunneling Protocol Se...
Running StateRepository State Repository Service
Running StorSvc Storage Service
Running SysMain SysMain
Running SystemEventsBroker System Events Broker
Running TextInputManage... Text Input Management Service
Running Themes Themes
Running TimeBrokerSvc Time Broker
Running TokenBroker Web Account Manager
Running TrkWks Distributed Link Tracking Client
Running UdkUserSvc_ab374af Udk User Service_ab374af
Running UnistoreSvc_ab3... User Data Storage_ab374af
Running UserDataSvc_ab3... User Data Access_ab374af
Running UserManager User Manager

```

Cybersecurity Lab #3 — Simulate Attacker Recon

Run:

whoami

hostname

ipconfig

net user

net localgroup administrators

These are common reconnaissance commands an attacker might use after gaining access to a system.

Your task is to explain:

- Who am I?
- What computer is this?
- What users exist?
- Who has admin rights?


```
PS C:\WINDOWS\System32> whoami
homeoffice\alyas
PS C:\WINDOWS\System32> hostname
Homeoffice
PS C:\WINDOWS\System32> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Unknown adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : lan
    IPv6 Address. . . . . : 2603:900b:300:61bc::1894
    IPv6 Address. . . . . : 2603:900b:300:61bc:8269:9f83:faa1:f7e6
    Temporary IPv6 Address. . . . . : 2603:900b:300:61bc:f953:a583:1ccf:efe8
    Link-local IPv6 Address . . . . . : fe80::3f39:e99d:48cc:5c30%4
    IPv4 Address. . . . . : 192.168.1.161
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::a255:1fff:fea6:7dad%4
                                192.168.1.1

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::fc00:b6b0:8bf1:9a05%10
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 4:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : lan

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
PS C:\WINDOWS\System32>
```

```
PS C:\WINDOWS\System32> net user

User accounts for \\HOMEOFFICE

-----
Administrator      ALYAS              DefaultAccount
Guest               WDAGUtilityAccount
The command completed successfully.

PS C:\WINDOWS\System32> net localgroup administrator
System error 1376 has occurred.

The specified local group does not exist.
```

Cybersecurity Lab #3- — Event Viewer Threat Hunting

In PowerShell:

```
Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4624} -MaxEvents 20
```

Look for:

Logon Type
Account Name
Time Created

This is real login investigation work.

, run:

```
Get-NetTCPConnection -State Listen | Sort-Object LocalPort
```

```
PS C:\WINDOWS\System32> Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4624} -MaxEvents 20
Get-WinEvent : No events were found that match the specified selection criteria.
At line:1 char:1
+ Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4624} -MaxEven ...
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (:) [Get-WinEvent], Exception
+ FullyQualifiedErrorId : NoMatchingEventsFound,Microsoft.PowerShell.Commands.GetWinEventCommand

PS C:\WINDOWS\System32> Get-NetTCPConnection -State Listen | Sort-Object LocalPort
```

LocalAddress	LocalPort	RemoteAddress	RemotePort	State	AppliedSetting	OwningProcess
::	135	::	0	Listen		1508
0.0.0.0	135	0.0.0.0	0	Listen		1508
192.168.1.161	139	0.0.0.0	0	Listen		4
192.168.56.1	139	0.0.0.0	0	Listen		4
::	445	::	0	Listen		4
127.0.0.1	2015	0.0.0.0	0	Listen		5284
0.0.0.0	5040	0.0.0.0	0	Listen		11120
0.0.0.0	6646	0.0.0.0	0	Listen		7148
::	7680	::	0	Listen		6936
127.0.0.1	15292	0.0.0.0	0	Listen		19340
::1	35783	::	0	Listen		3028
::	49664	::	0	Listen		1196
0.0.0.0	49664	0.0.0.0	0	Listen		1196
0.0.0.0	49665	0.0.0.0	0	Listen		1036
::	49665	::	0	Listen		1036
::	49666	::	0	Listen		836
0.0.0.0	49666	0.0.0.0	0	Listen		836
0.0.0.0	49667	0.0.0.0	0	Listen		2584
::	49667	::	0	Listen		2584
::	49668	::	0	Listen		4708
0.0.0.0	49668	0.0.0.0	0	Listen		4708
::	49673	::	0	Listen		1164
0.0.0.0	49673	0.0.0.0	0	Listen		1164
127.0.0.1	52363	0.0.0.0	0	Listen		25692

Lab 1: Find Unpatched Software

Check Windows Updates

Get-HotFix | Sort-Object InstalledOn -Descending

What you're looking for

- Latest update date
- Missing updates
- Systems not updated for months

```
PS C:\WINDOWS\System32> Get-HotFix | Sort-Object InstalledOn -Descending
```

Source	Description	HotFixID	InstalledBy	InstalledOn
HOMEOFFICE	Update	KB5089573	NT AUTHORITY\SYSTEM	5/28/2026 12:00:00 AM
HOMEOFFICE	Update	KB5092427	NT AUTHORITY\SYSTEM	5/28/2026 12:00:00 AM
HOMEOFFICE	Update	KB5092734	NT AUTHORITY\SYSTEM	5/27/2026 12:00:00 AM
HOMEOFFICE	Update	KB5054156	NT AUTHORITY\SYSTEM	2/23/2026 12:00:00 AM

Lab 2: Find Old Software

List Installed Programs

Get-Package | Sort-Object Name

Look for:

Java
Adobe Reader
Firefox
Chrome
7-Zip
VLC

Then check vendor websites for current versions.

What you're looking for

Old versions
Software no longer supported
End-of-life applications

```
PS C:\WINDOWS\System32> Get-Package | Sort-Object Name
```

Name	Version	Source	ProviderName
2026-01 .NET Framework Prev...			msu
2026-03 Security Update (KB...			msu
2026-03 Update (KB5085516) ...			msu
2026-03 Update (KB5086672) ...			msu
2026-04 .NET Framework Secu...			msu
2026-04 Preview Update (KB5...			msu
2026-04 Security Update (KB...			msu
2026-05 .NET Framework Prev...			msu
2026-05 .NET Framework Secu...			msu
2026-05 Preview Update (KB5...			msu
2026-05 Security Update (KB...			msu
9NKRJ35J9SDG-Microsoft.Wind...			msu
9PLJQ12FQ3CV-MicrosoftCorpo...			msu
Adobe Creative Cloud	6.9.1.1		Programs
Adobe Genuine Service	9.1.0.52		Programs
Adobe Photoshop 2023	24.3.0.376		Programs
Browser Extension	1.20.105.0	C:\Users\ALYAS\AppData\Roamin...	msi
Copilot	148.0.3967.96		Programs
DCS FB Marketplace Posting ...	1.0.0.108		Programs
DisplayLink - Display - 11...			msu
Epic Games Launcher	1.3.93.0	C:\Program Files (x86)\Epic G...	msi
Epic Games Launcher Prerequ...	1.0.0.0		msi
Epic Online Services	2.0.44.0		msi
ExpressVPN	10.0.9.2		Programs
ExpressVPN	10.0.9.2		msi
Google Chrome	149.0.7827.53		Programs
HP Audio Switch	1.0.211.0		msi
HP Connection Optimizer	2.0.19.0		Programs
HP Documentation	1.0.0.1		Programs
HP Inc. SoftwareComponent D...			msu
HP Inc. SoftwareComponent D...			msu
HP One Agent	1.3.214.7339		Programs
HP One Agent	1.3.214.7339	C:\Program Files\HP\HP One Ag...	msi
HP PC Hardware Diagnostics ...	9.1.0.0	C:\Windows\HP\	msi
Kraken Desktop	1.17.2		msi
McAfee LiveSafe	16.0 R43		Programs
Microsoft 365 - en-us	16.0.20026.20140		Programs
Microsoft Corporation Audio...			msu
Microsoft Corporation Drive...			msu
Microsoft Edge	149.0.4022.52		Programs
Microsoft GameInput	3.3.221.0		msi
Microsoft OneDrive	26.088.0510.0004		Programs
Microsoft OneNote - en-us	16.0.20026.20140		Programs
Microsoft Teams Meeting Add...	1.26.11802		msi
Microsoft Update Health Tools	5.72.0.0		msi
Microsoft Visual C++ 2012 R...	11.0.61030.0		Programs
Microsoft Visual C++ 2012 R...	11.0.61030.0		Programs
Microsoft Visual C++ 2012 x...	11.0.61030		msi
Microsoft Visual C++ 2012 x...	11.0.61030		msi
Microsoft Visual C++ 2012 x...	11.0.61030		msi
Microsoft Visual C++ 2012 x...	11.0.61030		msi
Microsoft Visual C++ 2013 R...	12.0.40664.0		Programs
Microsoft Visual C++ 2013 R...	12.0.40664.0		Programs
Microsoft Visual C++ 2013 x...	12.0.40664		msi
Microsoft Visual C++ 2013 x...	12.0.40664		msi
Microsoft Visual C++ 2013 x...	12.0.40664		msi

Lab 3: Find Listening Services

PowerShell

Get-NetTCPConnection -State Listen | Sort-Object LocalPort

Common ports

Port Service

135	RPC
445	SMB
3389	RDP
5985	WinRM
80	HTTP
443	HTTPS

What you're looking for

Services that shouldn't be exposed

Unexpected ports

Unknown software listening

```
PS C:\WINDOWS\System32> Get-NetTCPConnection -State Listen | Sort-Object LocalPort
```

LocalAddress	LocalPort	RemoteAddress	RemotePort	State	AppliedSetting	OwningProcess
::	135	::	0	Listen		1508
0.0.0.0	135	0.0.0.0	0	Listen		1508
192.168.1.161	139	0.0.0.0	0	Listen		4
192.168.56.1	139	0.0.0.0	0	Listen		4
::	445	::	0	Listen		4
127.0.0.1	2015	0.0.0.0	0	Listen		5284
0.0.0.0	5040	0.0.0.0	0	Listen		11120
0.0.0.0	6646	0.0.0.0	0	Listen		7148
::	7680	::	0	Listen		6936
127.0.0.1	15292	0.0.0.0	0	Listen		19340
::1	35783	::	0	Listen		3028
::	49664	::	0	Listen		1196
0.0.0.0	49664	0.0.0.0	0	Listen		1196
0.0.0.0	49665	0.0.0.0	0	Listen		1036
::	49665	::	0	Listen		1036
::	49666	::	0	Listen		836
0.0.0.0	49666	0.0.0.0	0	Listen		836
0.0.0.0	49667	0.0.0.0	0	Listen		2584
::	49667	::	0	Listen		2584
::	49668	::	0	Listen		4708
0.0.0.0	49668	0.0.0.0	0	Listen		4708
::	49673	::	0	Listen		1164
0.0.0.0	49673	0.0.0.0	0	Listen		1164
127.0.0.1	52363	0.0.0.0	0	Listen		25692

Lab 4: Find Web Servers on Your Network

From Kali:

```
nmap -sV 192.168.1.0/24
```

What you're looking for

Examples:

Apache 2.4.48

nginx 1.18

IIS 10

The version number matters.

Then research

Search:

Apache 2.4.48 vulnerabilities

nginx 1.18 CVEs

SOC Skill

Service Enumeration

Version Identification

```
PS C:\WINDOWS\System32> nmap -sV 192.168.1.0/24
nmap: The term 'nmap' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ nmap -sV 192.168.1.0/24
~
+ CategoryInfo          : ObjectNotFound: (nmap:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException
```

Lab 5: Identify VPN Software

On Windows:

```
Get-Service | Where-Object {
$_.DisplayName -match "VPN|Cisco|Fortinet|GlobalProtect|AnyConnect"
}
```

Look for

Cisco AnyConnect

FortiClient

GlobalProtect

OpenVPN

WireGuard

Why?

Security teams track:

Outdated VPN clients

Vulnerable VPN appliances

Unused VPN software

```
PS C:\WINDOWS\System32> Get-Service | Where-Object {  
>> $_.DisplayName -match "VPN|Cisco|Fortinet|GlobalProtect|AnyConnect"  
>> }
```

Status	Name	DisplayName
Running	ExpressVPNService	ExpressVPN Service

Lab 6: Check SMB Configuration

PowerShell:

Get-SmbServerConfiguration

What you're looking for

Especially:

EnableSMB1Protocol

Good

False

Bad

True

SMBv1 is old and insecure.


```
PS C:\WINDOWS\System32> Get-SmbServerConfiguration
```

```
AnnounceComment           :  
AnnounceServer            : False  
AsynchronousCredits       : 64  
AuditClientCertificateAccess : False  
AuditClientDoesNotSupportEncryption : False  
AuditClientDoesNotSupportSigning : False  
AuditInsecureGuestLogon   : False  
AuditSmb1Access           : False  
AutoDisconnectTimeoutInMinutesV1 : 15  
AutoDisconnectTimeoutInSecondsV2 : 900  
AutoShareServer           : True  
AutoShareWorkstation      : True  
CachedOpenLimit           : 10  
DisableCompression        : False  
DisableSmbEncryptionOnSecureConnection : True  
DurableHandleV2TimeoutInSeconds : 180  
EnableAuthenticateUserSharing : False  
EnableAuthRateLimiter     : True  
EnableDirectoryHandleLeasing : True  
EnableDownlevelTimewarp   : False  
EnableForcedLogoff        : True  
EnableLeasing              : True  
EnableMailslots           : False  
EnableMultiChannel        : True  
EnableOplocks              : True  
EnableSecuritySignature   : False  
EnableSMB1Protocol        : False  
EnableSMB2Protocol        : True  
EnableSMBQUIC              : True  
EnableStrictNameChecking  : True  
EncryptData                : False  
EncryptionCiphers         : AES_128_GCM, AES_128_CCM, AES_256_GCM, AES_256_CCM  
InvalidAuthenticationDelayTimeInMs : 2000  
IrpStackSize               : 15  
KeepAliveTime              : 2  
MaxChannelPerSession      : 32  
MaxMpxCount                : 50  
MaxSessionPerConnection   : 16384  
MaxThreadsPerQueue        : 20  
MaxWorkItems               : 1  
NullSessionPipes          :  
NullSessionShares         :  
OplockBreakWait           : 35  
PendingClientTimeoutInSeconds : 120  
RejectUnencryptedAccess   : True  
RequestCompression        : False  
RequireSecuritySignature   : False  
RestrictNamedpipeAccessViaQuic : True  
ServerHidden               : True  
Smb2CreditsMax            : 2048  
Smb2CreditsMin            : 128  
Smb2DialectMax            : None  
Smb2DialectMin            : None  
SmbServerNameHardeningLevel : 0  
TreatHostAsStableStorage  : False  
ValidateAliasNotCircular   : True  
ValidateShareScope        : True  
ValidateShareScopeNotAliased : True
```

Lab 7: Security Analyst Network Audit

In Kali.

Run:

```
nmap -sn 192.168.1.0/24
```

Then:

```
nmap -sV 192.168.1.83
```

(your Windows machine)

What you're looking for

Open ports

Service versions

Unexpected services

```
PS C:\WINDOWS\System32> nmap -sn 192.168.1.0/24
nmap : The term 'nmap' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ nmap -sn 192.168.1.0/24
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (nmap:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\WINDOWS\System32> nmap -sn 192.168.1.0/24
nmap : The term 'nmap' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ nmap -sn 192.168.1.0/24
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (nmap:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\WINDOWS\System32> nmap -sV 192.168.1.83
nmap : The term 'nmap' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ nmap -sV 192.168.1.83
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (nmap:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\WINDOWS\System32>
```

```
(kali㉿kali)-[~]  
$ nmap -sV 192.168.1.83  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-06 16:04 -0400  
setup_target: failed to determine route to 192.168.1.83  
WARNING: No targets were specified, so 0 hosts scanned.  
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.18 seconds  
  
(kali㉿kali)-[~]  
$
```

Get-SmbServerConfiguration | Select EnableSMB1Protocol

Get-HotFix | Select -First 10

Get-Service | Where-Object {\$_.Status -eq "Running"} | Select -First 20

```

PS C:\WINDOWS\System32>
PS C:\WINDOWS\System32> Get-Service | Where-Object {$_.Status -eq "Running"} | Select -First 20

```

Status	Name	DisplayName
Running	AdobeUpdateService	AdobeUpdateService
Running	AGMSvc	Adobe Genuine Software Monitor Service
Running	AMD Crash Defen...	AMD Crash Defender Service
Running	AMD External Ev...	AMD External Events Utility
Running	Appinfo	Application Information
Running	AppXSvc	AppX Deployment Service (AppXSVC)
Running	AudioEndpointBu...	Windows Audio Endpoint Builder
Running	Audiosrv	Windows Audio
Running	BFE	Base Filtering Engine
Running	BITS	Background Intelligent Transfer Ser...
Running	BluetoothUserSe...	Bluetooth User Support Service_ab374af
Running	BrokerInfrastru...	Background Tasks Infrastructure Ser...
Running	BTAGService	Bluetooth Audio Gateway Service
Running	BthAvctpSvc	AVCTP service
Running	bthserv	Bluetooth Support Service
Running	camsvc	Capability Access Manager Service
Running	cbdhsvc_ab374af	Clipboard User Service_ab374af
Running	CDPSvc	Connected Devices Platform Service
Running	CDPUserSvc_ab374af	Connected Devices Platform User Ser...
Running	ClickToRunSvc	Microsoft Office Click-to-Run Service

```

PS C:\WINDOWS\System32> Get-SmbServerConfiguration | Select EnableSMB1Protocol

```

EnableSMB1Protocol
False

```

PS C:\WINDOWS\System32>
PS C:\WINDOWS\System32> Get-HotFix | Select -First 10

```

Source	Description	HotFixID	InstalledBy	InstalledOn
HOMEOFFICE	Update	KB5092427	NT AUTHORITY\SYSTEM	5/28/2026 12:00:00 AM
HOMEOFFICE	Update	KB5054156	NT AUTHORITY\SYSTEM	2/23/2026 12:00:00 AM
HOMEOFFICE	Update	KB5089573	NT AUTHORITY\SYSTEM	5/28/2026 12:00:00 AM
HOMEOFFICE	Update	KB5092734	NT AUTHORITY\SYSTEM	5/27/2026 12:00:00 AM

Excellent work. You're starting to think like a SOC analyst now.

Lab Analysis

SMBv1 Check

EnableSMB1Protocol = False

This is good.

Finding:

- SMBv1 disabled
- Not vulnerable to many older SMB attacks such as WannaCry-style SMBv1 exploitation

Risk Level: Low

Patch Level Check

Latest updates:

KB5092427

KB5054156

KB5089573

KB5092734

Installed:

May 27-28, 2026

Finding:

My recently patched.

As a SOC analyst I would write:

Workstation appears current on Microsoft security updates. No evidence of significantly delayed patching.

Risk Level: Low

** Running Services Review**

Most of these are expected:

Service	Normal?
BFE	
BITS	
AppInfo	

AudioSrv 

ClickToRunSvc 

Bluetooth Services 

AdobeUpdateService 

Nothing jumps out as obviously malicious.

Lab 1

Open PowerShell as Administrator:

Get-Process | Sort-Object CPU -Descending | Select -First 20

Look for:

powershell

cmd

wscript

cscript

mshta

rundll32

regsvr32

These are commonly abused by attackers.

```
PS C:\WINDOWS\System32> Get-Process | Sort-Object CPU -Descending | Select -First 20
```

Handles	NPM(K)	PM(K)	WS(K)	CPU(s)	Id	SI	ProcessName
435	40	480076	461256	2,035.63	30900	5	chrome
1165	158	217632	18772	1,513.39	35604	5	VirtualBoxVM
1288	48	628536	23032	1,082.39	308	5	XboxPcApp
1905	54	414068	133704	654.48	15268	5	chrome
2364	80	134244	130612	348.55	24224	5	chrome
623	73	352320	307476	218.23	34784	5	chrome
1270	143	191840	15092	216.02	12764	5	EpicGamesLauncher
3609	105	142484	106004	113.27	10576	5	explorer
479	34	26044	28752	104.78	11472	5	chrome
1241	43	39984	16672	100.70	31596	5	msedgewebview2
2334	102	997792	219968	92.92	22660	5	SnippingTool
338	45	221348	19024	92.84	15964	5	msedgewebview2
1287	85	82532	15568	60.41	19340	5	Adobe Desktop Service
573	32	15148	14912	60.17	34944	5	uihost
487	45	221232	152152	46.53	31048	5	chrome
498	36	29668	17880	34.52	10496	5	node
497	28	65332	12844	33.25	25036	5	VirtualBox
918	44	33304	13380	32.94	17392	5	Creative Cloud
273	14	4400	5964	27.69	2272	5	conhost
519	31	37176	2556	25.00	24720	5	DesktopExtension

Lab 2 — Find Scheduled Tasks

Attackers often create persistence.

Run:

```
Get-ScheduledTask | Where-Object {$_.State -eq "Ready"} | Select -First 20
```

What you're looking for

Suspicious names like:

Windows Update Helper

ChromeUpdater

Adobe Security Service

that don't belong.

```
PS C:\WINDOWS\System32> Get-ScheduledTask | Where-Object {$_.State -eq "Ready"} | Select -First 20
```

TaskPath	TaskName	State
	Adobe-Genuine-Software-Integri...	Ready
	AdobeGCInvoker-1.0	Ready
	HPSupportTool	Ready
	McAfeeLogon	Ready
	OneDrive Reporting Task-S-1-5-...	Ready
	OneDrive Standalone Update Tas...	Ready
	OneDrive Startup Task-S-1-5-21...	Ready
GoogleUserPEH\	RunPlatformExperienceHelper_Daily	Ready
GoogleUserPEH\	RunPlatformExperienceHelper_Me...	Ready
Hewlett-Packard\HP Support Assistant\	HP Support Assistant Update No...	Ready
Hewlett-Packard\HP Support Assistant\	WarrantyChecker	Ready
Hewlett-Packard\HP Support Assistant\	WarrantyChecker_DeviceScan	Ready
HP\HP Print Scan Doctor\	Printer Health Monitor	Ready
HP\HP Print Scan Doctor\	Printer Health Monitor Logon	Ready
Microsoft\Office\	Office Actions Server	Ready
Microsoft\Office\	Office Automatic Updates 2.0	Ready
Microsoft\Office\	Office Background Push Mainten...	Ready
Microsoft\Office\	Office ClickToRun Service Monitor	Ready
Microsoft\Office\	Office Feature Updates	Ready
Microsoft\Office\	Office Feature Updates Logon	Ready

Lab 3 — Find Startup Programs

Run:

```
Get-CimInstance Win32_StartupCommand
```

What attackers abuse

Programs that start automatically when Windows boots.

```
PS C:\WINDOWS\System32> Get-CimInstance Win32_StartupCommand
```

Command	User	Caption
C:\Program Files\HP\HP System Event Utility\Host Launcher\HpseuHostLauncher.exe	HOMEOFFICE\ALYAS	HPSEU_Host_Launcher
"C:\Users\ALYAS\AppData\Local\Microsoft\OneDrive\OneDrive.exe" /background	HOMEOFFICE\ALYAS	OneDrive
A&B&C	HOMEOFFICE\ALYAS	
"C:\Program Files (x86)\Epic Games\Launcher\Portal\Binaries\Win64\EpicGamesLauncher.exe" -silent -launchcontext=boot	HOMEOFFICE\ALYAS	EpicGamesLauncher
"C:\Users\ALYAS\AppData\Local\Microsoft\WindowsApps\MSTeams_8wekyb3d8bbwe\ms-teams.exe" msteams:system-initiated	HOMEOFFICE\ALYAS	Teams
"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --no-startup-window --win-session-start	HOMEOFFICE\ALYAS	MicrosoftEdgeAutoLaunch_CED900BA7...
%windir%\system32\SecurityHealthSystray.exe	Public	SecurityHealth
"C:\Program Files (x86)\Common Files\Adobe\AdobeGCClient\AGCInvokerUtility.exe"	Public	AdobeGCInvoker-1.0

Lab 4 — Map Network Connections to Processes

Run:

Get-NetTCPConnection -State Established |

Select LocalAddress,RemoteAddress,RemotePort,OwningProcess

Then take one PID and investigate it:

Get-Process -Id PIDNUMBER

```
PS C:\WINDOWS\System32> Get-CimInstance Win32_StartupCommand
Command
-----
C:\Program Files\HP\HP System Event Utility\Host Launcher\HpseuHostLauncher.exe
'C:\Users\ALYAS\AppData\Local\Microsoft\OneDrive\OneDrive.exe' /background
C:\Program Files (x86)\Epic Games\Launcher\Portal\Binaries\Win64\EpicGamesLauncher.exe" -silent -l
'C:\Users\ALYAS\AppData\Local\Microsoft\WindowsApps\MSTeams_8wekyb3d8bbwe\ms-teams.exe" msteams:sys
'C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --no-startup-window --win-session-st
%windir%\system32\SecurityHealthSystray.exe
'C:\Program Files (x86)\Common Files\Adobe\AdobeGCCClient\AGCInvokerUtility.exe"

PS C:\WINDOWS\System32> Get-NetTCPConnection -State Established |
>> Select LocalAddress,RemoteAddress,RemotePort,OwningProcess

LocalAddress                                RemoteAddress                                RemotePort OwningProcess
-----
2603:900b:300:61bc:6cd8:816d:86f0:727c 2607:f8b0:4008:806::200e                      443        11472
2603:900b:300:61bc:6cd8:816d:86f0:727c 2607:f8b0:400c:c05::bc                        5228       11472
2603:900b:300:61bc:6cd8:816d:86f0:727c 2a06:98c1:3108::ac40:94eb                     443        11472
2603:900b:300:61bc:6cd8:816d:86f0:727c 2606:4700:4406::6812:2bcc                     443        11472
2603:900b:300:61bc:6cd8:816d:86f0:727c 2607:f8b0:4008:817::2003                      443        11472
192.168.1.161                          192.168.1.120                               8009       11472
192.168.1.161                          172.183.7.192                               443        5376
127.0.0.1                              127.0.0.1                                   52363      5284
127.0.0.1                              127.0.0.1                                   52365      25692
192.168.1.161                          100.55.19.214                               443        27064
```

Lab 5 — Search for Encoded PowerShell

Attackers love Base64 encoded PowerShell.

Search PowerShell logs:

```
Get-WinEvent -LogName "Microsoft-Windows-PowerShell/Operational" |
Where-Object {$_.Message -match "EncodedCommand"} |
Select -First 20
```

What you're looking for

powershell.exe -EncodedCommand

This is a major red flag.

```
>> Select -First 20
PS C:\WINDOWS\System32> Get-WinEvent -LogName "Microsoft-Windows-PowerShell/Operational" |
>> Where-Object {$_.Message -match "EncodedCommand"} |
>> Select -First 20

ProviderName: Microsoft-Windows-PowerShell

TimeCreated              Id LevelDisplayName Message
-----
6/6/2026 10:09:15 PM      4104 Warning      Creating Scriptblock text (1 of 1):...
6/6/2026 10:09:15 PM      4104 Warning      Creating Scriptblock text (1 of 1):...
```

Threat Hunter Lab

Run:

Get-WinEvent -LogName Security -MaxEvents 200 |

Where-Object {\$_.Id -eq 4688}

4688 = Process Creation

Look for:

powershell.exe

cmd.exe

wscript.exe

mshta.exe

rundll32.exe

These are among the most common tools attackers abuse after gaining access.

```
PS C:\WINDOWS\System32> Get-WinEvent -LogName "Microsoft-Windows-PowerShell/Operational" |
> Where-Object {$_.Message -match "EncodedCommand"} |
> Select -First 20
PS C:\WINDOWS\System32> Get-WinEvent -LogName "Microsoft-Windows-PowerShell/Operational" |
> Where-Object {$_.Message -match "EncodedCommand"} |
> Select -First 20

ProviderName: Microsoft-Windows-PowerShell

TimeCreated              Id LevelDisplayName Message
-----
6/6/2026 10:09:15 PM      4104 Warning      Creating Scriptblock text (1 of 1):...
6/6/2026 10:09:15 PM      4104 Warning      Creating Scriptblock text (1 of 1):...

PS C:\WINDOWS\System32> Get-WinEvent -LogName Security -MaxEvents 200 |
> Where-Object {$_.Id -eq 4688}
Get-WinEvent : Attempted to perform an unauthorized operation.
at line:1 char:1
Get-WinEvent -LogName Security -MaxEvents 200 |
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [Get-WinEvent], UnauthorizedAccessException
+ FullyQualifiedErrorId : System.UnauthorizedAccessException,Microsoft.PowerShell.Commands.GetWinEventCommand
```

PowerShell Log Result

You found:

Event ID 4104

Creating Scriptblock text (1 of 1)

That means **PowerShell Script Block Logging** is enabled and recording PowerShell activity.

What you did **not** find:

EncodedCommand

That's good.

If an attacker launched something like:

powershell.exe -EncodedCommand SQBFaFgA...

it would often show up here.

Finding:

- No obvious Base64-encoded PowerShell activity detected.

Security Log Error

You got:

Attempted to perform an unauthorized operation

This usually means PowerShell doesn't have permission to read the Security log.

Open **PowerShell as Administrator** (Run as Administrator) and try:

```
Get-WinEvent -LogName Security -MaxEvents 20
```

Challenge #1

Let's investigate active network connections.

Run:

```
Get-NetTCPConnection -State Established |  
Select LocalAddress,RemoteAddress,RemotePort,OwningProcess
```

Then pick one PID and investigate it:

```
Get-Process -Id PIDNUMBER
```

Example:

```
Get-Process -Id 12804
```

Question

Can you identify:

- Chrome?
- Edge?
- OneDrive?
- Adobe?
- Something unexpected?

This is real threat hunting.

```
PS C:\Users\ALYAS> Get-NetTCPConnection -State Established |
>> Select LocalAddress,RemoteAddress,RemotePort,OwningProcess
```

LocalAddress	RemoteAddress	RemotePort	OwningProcess
2603:900b:300:61bc:6cd8:816d:86f0:727c	2607:f8b0:400c:c13::bc	5228	11472
2603:900b:300:61bc:6cd8:816d:86f0:727c	2607:f8b0:4008:806::200e	443	11472
2603:900b:300:61bc:6cd8:816d:86f0:727c	2620:1ec:33:1::11	443	30956
2603:900b:300:61bc:6cd8:816d:86f0:727c	2607:f8b0:4023:3801::5e	443	11472
2603:900b:300:61bc:6cd8:816d:86f0:727c	2607:f8b0:4023:3801::5e	443	11472
2603:900b:300:61bc:6cd8:816d:86f0:727c	2a06:98c1:3108::ac40:94eb	443	11472
2603:900b:300:61bc:6cd8:816d:86f0:727c	2620:1ec:33:1::11	443	30956
2603:900b:300:61bc:6cd8:816d:86f0:727c	2607:f8b0:4023:3801::5e	443	11472
192.168.1.161	20.190.157.4	443	35356
192.168.1.161	20.184.175.21	443	4568
192.168.1.161	20.42.65.91	443	6896
192.168.1.161	20.190.157.4	443	35356
192.168.1.161	142.251.214.227	443	11472
127.0.0.1	127.0.0.1	52363	5284
127.0.0.1	127.0.0.1	52365	25692
192.168.1.161	192.168.1.120	8009	11472
192.168.1.161	172.183.7.192	443	5376

```
PS C:\Users\ALYAS> Get-Process -Id 5376
```

Handles	NPM(K)	PM(K)	WS(K)	CPU(s)	Id	SI	ProcessName
501	21	5552	2732		5376	0	svchost

```
PS C:\Users\ALYAS> Get-Process -Id 11472
```

Handles	NPM(K)	PM(K)	WS(K)	CPU(s)	Id	SI	ProcessName
505	38	25888	30856	131.59	11472	5	chrome